

Avinash Chowdary Kongara

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[Portfolio](#)

SUMMARY

Full-Stack Software Engineer with 4 years of experience designing and shipping scalable, production-grade applications across Java, Spring Boot, React, and Node.js. Specialized in event-driven microservices with Vert.x, REST API development, and TypeScript-based UI component systems. Experienced in end-to-end cloud delivery from Kubernetes deployments and CI/CD pipelines to DevSecOps practices with a track record of optimizing system performance and reducing delivery friction in Agile environments.

EDUCATION

Ph.D. in IT (spec. in Artificial Intelligence) *	University of the Cumberlands	Jan 2026 – Current
Master of Science, Computer Science	University of Missouri–Kansas City	Aug 2021 - Dec 2022

SKILLS

Languages: Java, JavaScript, TypeScript, Python, SQL

Frontend: React, React Native, Redux, HTML5, CSS3, Material UI, Bootstrap, Ant Design

Backend: Vert.x, Spring Boot, Node.js, FastAPI

Cloud & DevOps: GCP, Azure DevOps, Kubernetes, Docker, Rancher, GitHub Actions, Jenkins

Testing: Jest, Mocha, Cypress, JUnit, Mockito, React Testing Library

Databases: PostgreSQL, MySQL, DB2, MS SQL

Tools & Platforms: Backstage.io, Jira, Rally, Git, Terraform, HashiCorp Vault, SonarQube

Concepts: Microservices, Event-Driven Architecture, SDLC, CI/CD, UI/UX, REST APIs

PROFESSIONAL EXPERIENCE

Software AI Engineer

American Express – Phoenix, AZ

Jan 2025 – Present

- Worked on developing an Governance application for automation of Gen-AI/ML model analysis project that was built on the Vert.x framework (one-data) with Java and one-app (React) for the UI.
- Developed Elastic search capability in the application, where indexing happening on regular basis for user quick search results in entire application.
- Built scalable, event-driven microservices using Vert.x and Java, enabling high-performance asynchronous workloads.
- Developed RESTful APIs using the Vert.x toolkit and Java, supporting asynchronous data flows and integrating with internal/external services.
- Designed and optimized complex PostgreSQL queries, views, and stored procedures, improving performance by 30%.
- Automated data transformation and orchestration flows using ACE integration patterns and BPMN workflows.

- Performed schema design, indexing, and performance tuning for PostgreSQL-backed applications.
 - Participated in full SDLC including requirements gathering, technical design, development, testing, and deployment.
 - Leveraged Maven, Git, and CI/CD pipelines for automated build, test, and deployment processes.
 - Collaborated in Agile ceremonies including grooming, planning, and retrospectives to improve team delivery.
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Software Engineer (Cybersecurity)

American Airlines – Fort Worth, TX

Sep 2022 – Dec 2024

- Worked on advanced user interface design (designing mocks), application deployment (software development), and web development (UI/UX) using React v18, JavaScript, FastAPI, Python, React Native TypeScript (v5), Node.js (v17/20/22), ES6, Next.js, HTML5, Next.js, JSX, AJAX, Ant Design, CSS, and REST API development using Java.
 - Built reusable React component libraries used across multiple micro-frontends, increasing dev efficiency and UI consistency.
 - Implemented micro-frontend architecture (Module Federation, single-spa) for independent deployment of feature apps.
 - Developed Backstage.io developer portal plugins using React, TypeScript, and Python scripting.
 - Integrated HashiCorp Vault with Kubernetes for dynamic secrets management, improving application security posture. Rancher for cluster management
 - Automated CI/CD workflows using GitHub Actions, optimizing build, test, and deployment pipelines.
 - Deployed applications in Azure shared Kubernetes clusters using Docker.
 - Utilized advanced ES6+ and React Hooks to build maintainable, scalable UI applications.
 - Worked extensively with DB2, PostgreSQL, and MS SQL databases for backend integration and performance tuning.
 - Collaborated closely with UX teams and product owners to deliver user-centered design improvements.
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Software Developer (Intern)

Airports Authority of India – Hyderabad, India

May 2020 – oct 2020

- Built interactive UI screens using HTML5, CSS, TypeScript, RxJS, AJAX, and React for an internal aviation analytics system.
 - Implemented Node.js for server-side rendering and backend logic.
 - Integrated third-party APIs (Axios, GraphQL) to enable dynamic, real-time data updates.
 - Developed reusable React patterns (HOCs, Render Props) to improve modularity and code reuse.
 - Utilized Jenkins for CI/CD automation, reducing deployment time and improving release reliability.
 - Implemented advanced Java programming for backend integrations (multithreading, collections, exception handling).
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